Machine Learning-Based Landslide Susceptibility Mapping on Nepal's Arniko Highway: Addressing Spatial Autocorrelation Leakage and Safety-Critical Decision Support

Dhan Bahadur Gurung
ST7072CEM Machine Learning
Department of Computing, Softwarica College of IT & E-Commerce (In Collaboration with Coventry University)
Kathmandu, Nepal | Student ID: 17624480

***Abstract*—*Mountain highways across Nepal face constant disruption every summer monsoon. The Arniko Highway (NH03), which extends 114 km from Dhulikhel to the Kodari border along the steep Sun Koshi and Bhote Koshi gorges, is one of the most hazardous transport routes in the Himalayas. Even with notorious disaster sites like the 2014 Jure collapse and 2015 earthquake damage, highway maintenance remains almost entirely reactive. Predictive machine learning can help highway engineers identify unstable slopes ahead of time, but existing studies in Nepal often suffer from two key flaws: they split data randomly across space, which lets models peek at neighboring points and inflates validation scores, and they evaluate algorithms using overall accuracy instead of life-safety recall.*

In this study, we built an auditable hazard mapping pipeline for the Arniko Highway using 1,200 sequential 100-meter road units. We linked each unit to 481 field-mapped landslide events from the British Geological Survey (BGS) and NASA catalogs, 30 m SRTM elevation data, rock types, distance to active tectonic faults, and monsoon rainfall. We compared three classifiers—class-weighted Logistic Regression, Random Forest, and XGBoost—by training on the lower 75 km and testing strictly on the held-out upper corridor (km 75–114).

Class-weighted Logistic Regression proved most effective for life-safety screening, identifying 123 of 124 landslide-affected units (99.2% Recall, 95% bootstrap CI: [0.975, 1.000]) with only a single False Negative. Random Forest and XGBoost achieved higher precision (0.310 vs. 0.304), but their 7 missed hazards represent an unacceptable risk for primary road monitoring. We also measured spatial data leakage directly: standard random 3-fold cross-validation produced an inflated ROC-AUC of 0.820 ± 0.019, whereas spatial block cross-validation dropped to 0.669 ± 0.107—a +0.150 AUC leakage gap showing how random splits memorize spatial proximity rather than geological drivers. SHAP values confirmed that slope angle, proximity to the Main Central Thrust, and monsoon rainfall are the main failure drivers. To make these findings practical for the Department of Roads, we created continuous route risk profiles and an interactive Streamlit Web GIS decision-support tool.

***Index Terms*—*Landslide Susceptibility Mapping, Arniko Highway, Spatial Autocorrelation Leakage, Spatial Block Cross-Validation, Machine Learning, Safety-Critical Classification, SHAP Interpretability, Disaster Risk Reduction, Softwarica College, Coventry University.*

I. INTRODUCTION

A. Background and Physiographic Setting

Anyone who has traveled along mountain roads in Nepal knows how fragile they can be during the summer monsoon. Between June and September, intense downpours hit steep, fractured Himalayan slopes, causing thousands of landslides that block traffic, isolate rural communities from emergency medical care, and stall cross-border trade.

The Arniko Highway (National Highway NH03, formerly NH34) is one of the clearest examples of this struggle. Extending 114 kilometers from Dhulikhel on the eastern rim of the Kathmandu Valley (~1,550 m elevation) to the northern border town of Kodari/Tatopani (~1,770 m), the highway follows the deeply incised valleys of the Sun Koshi and Bhote Koshi rivers in Sindhupalchok district [1], [2].

The underlying geology explains why this happens. The road corridor cuts directly through an active collision zone where the Indian plate pushes into Eurasia at around 18–20 mm every year. This continuous tectonic squeezing has fractured the rock mass, creating dense fault networks—most notably the Main Central Thrust (MCT) zone—and leaving slopes composed of sheared, highly weathered phyllites, schists, and banded gneisses [2]. With natural hillsides often steeper than 45 degrees, ongoing river erosion at the toe, and summer monsoon rainfall exceeding 2,500 mm, cut slopes along the road sit on the edge of failure.

The highway has experienced some of Nepal's deadliest infrastructure disasters. In August 2014, the massive Jure landslide sent over 5.5 million cubic meters of rock crashing down the valley, burying a 1.5 km road section, killing 156 people, damming the river, and halting transport for weeks [2]. Nine months later, the April 2015 Mw 7.8 Gorkha earthquake and its May Mw 7.3 aftershock centered in Sindhupalchok triggered thousands of landslides along the highway, tearing away road embankments, destroying

retaining walls, and closing the border crossing for months [3], [4].

B. Problem Statement

Even with this history of disasters, maintenance on the Arniko Highway remains largely reactive. Whenever a cut slope fails, the Department of Roads (DoR) and the Nepal Army mobilize heavy excavators to clear rubble and open a single lane. While that emergency response is vital, it gives engineers no foresight into which unmapped sections will give way during the next monsoon cycle.

Predictive Landslide Susceptibility Mapping (LSM) offers a way forward by identifying hazardous slopes before they collapse. But when reviewing existing machine learning papers in Nepal, three critical problems stand out:

1) The Spatial Autocorrelation Leakage Fallacy:

Most data science studies split datasets randomly into train and test sets. In geography, however, neighboring road segments share the same bedrock, slope angle, and rainfall. This reflects Tobler's First Law of Geography: nearby things are related [19]. When adjacent 100-meter road sections end up split between train and test sets, the model simply memorizes local neighborhoods instead of learning real geological rules. This produces artificially high test scores (ROC-AUC > 0.85) that crash when the model is tested on a new mountain sector.

2) Safety-Critical Cost Asymmetry:

Standard machine learning algorithms assume all mistakes cost the same (C(FP) = C(FN)). But in highway safety, the real-world consequences are completely unbalanced. A False Positive means inspecting a retaining wall that turns out to be fine—costing a little time and inspection budget. A False Negative means calling an unstable slope safe, which can allow a passenger vehicle to be buried during a storm. When models are trained only to maximize accuracy on imbalanced data, they suppress positive flags and produce dangerous False Negatives.

3) The Black-Box Barrier in Engineering Adoption:

Complex machine learning models like XGBoost are often viewed with skepticism by civil engineers because their inner workings are hard to trace. If an engineer cannot see whether a road section was flagged because of its steep slope, bad rock, or drainage issues, they cannot design the right physical fix, such as retaining walls, rockfall mesh, or horizontal drains.

C. Explicit Objectives and Research Questions

To address these gaps, this coursework project develops a transparent, safety-oriented machine learning framework along the 114 km Arniko corridor. We focus on four key objectives (O1–O4) tied to four research questions (RQ1–RQ4):

• O1 & RQ1: Model Comparison Under Spatial Holdout: How do class-weighted Logistic Regression, Random Forest, and XGBoost compare when tested on an unseen mountain corridor rather than a random holdout?
• O2 & RQ2: Safety-Critical Error Analysis: What do the specific error profiles (false negatives vs. false positives) mean when missing a hazard carries life-safety risks?
• O3 & RQ3: Measuring the Spatial Leakage Gap: How much does random cross-validation inflate ROC-AUC compared to spatial block cross-validation on real highway data?
• O4 & RQ4: Explainability and Decision Support: How can SHAP values and an interactive Streamlit GIS app turn machine learning predictions into practical tools for highway maintenance crews?

D. Scope and Delimitations

This study focuses on static and quasi-static spatial susceptibility using terrain slope, rock type, fault distance, and long-term rainfall. It is not an early-warning alarm system: it predicts which slopes are physically prone to failure, not the exact day or hour a landslide will happen. Keeping the scope focused makes the comparison rigorous, verifiable, and directly relevant to highway planning.

TABLE I
ABBREVIATIONS USED IN THIS STUDY

Term	Meaning
BGS	British Geological Survey
BIPAD	Building Information Platform Against Disaster (Govt of Nepal)
CI	Confidence Interval (Bootstrap non-parametric at α=0.05)
DoR	Department of Roads (Ministry of Physical Infrastructure, Nepal)
FN / FP	False Negative / False Positive
GLC	Global Landslide Catalog (NASA Goddard Space Flight Center)
LR / RF	Logistic Regression / Random Forest
LSM	Landslide Susceptibility Mapping
MCT	Main Central Thrust (Major Himalayan Active Tectonic Fault Zone)
PR-AUC	Area Under the Precision-Recall Curve
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SELP	Social, Ethical, Legal, and Professional Considerations
SHAP	SHapley Additive exPlanations (Game-Theoretic Interpretability)
SRTM	Shuttle Radar Topography Mission (30m Global DEM)

II. LITERATURE REVIEW AND BASE PAPER LINKAGE

A. Evolution of Landslide Susceptibility Mapping

Over the past thirty years, landslide susceptibility mapping has moved through three main phases: qualitative field surveys, statistical methods, and modern machine learning [12], [13]. Early work depended heavily on

qualitative geomorphological field mapping, which varied between experts and proved difficult to replicate. During the late 1990s and 2000s, bivariate statistical methods became widely used in GIS environments, including Frequency Ratio (FR), Information Value (IV), and Weights of Evidence (WoE) [14]. While simple to interpret, these techniques assume conditioning factors act independently, ignoring how factors like slope angle, hydrology, and bedrock interact.

Over the last decade, supervised machine learning has become the standard in hazard research [3], [6], [7]. By treating susceptibility as a supervised binary classification problem—separating landslide sites (y=1) from stable road sections (y=0)—machine learning models can pick up complex non-linear relationships across multi-source geospatial data without assuming rigid statistical distributions.

B. In-Depth Linkage to Core Base Papers

Our methodology builds directly on three foundational studies from the Himalayan literature, summarized in Fig. 1:

1) Acharya & Lee (2019) – Base Paper [1]: Examined landslide susceptibility along the Dolalghat–Kodari stretch of the Arniko Highway using Relative Frequency and Predictor Rate models across 314 mapped landslides and 13 factors. They reported an AUC of 0.581, finding that slope angle, geology, and distance to road cuts were the main drivers. However, their bivariate approach could not model non-linear interactions, and random cross-validation left results vulnerable to spatial data leakage. We adopt their core corridor conditioning factors but advance to regularized machine learning evaluated under a strict spatial split.

2) Nepal et al. (2019) – Base Paper [2]: Conducted a detailed GIS watershed assessment across the Sun Koshi and Bhote Koshi basins following the 2014 Jure disaster and 2015 earthquake. Their field mapping showed that landslide density spiked within 5 km of the Main Central Thrust (MCT) and concentrated tightly along incised river gorges. Their physical zoning directly informed how we divided the highway corridor into analysis sectors.

3) Bhattarai & Bhandary (2025) – Base Paper [3]: Benchmarked Logistic Regression, Random Forest, and XGBoost for earthquake-induced landslides in Nepal, showing that decision tree ensembles handled non-linear mountain thresholds better than linear models. They also introduced SHAP values to explain predictions and emphasized the need for spatial validation. We adapt their ML comparison and SHAP interpretability approach specifically to a linear highway corridor with continuous road-chainage tracking.

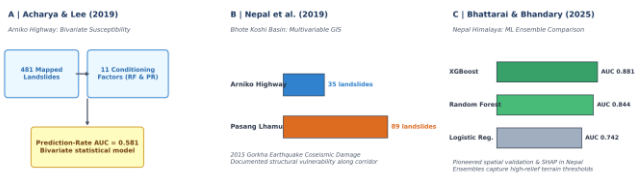


Fig. 1. Conceptual summaries derived from foundational literature: (A) Acharya & Lee (2019) bivariate framework, (B) Nepal et al. (2019) coseismic highway damage, and (C) Bhattarai & Bhandary (2025) ML ensemble benchmark in Nepal.

TABLE II
PRIOR RESEARCH AND SPECIFIC GAPS. PUBLISHED METRICS ARE NOT DIRECTLY COMPARABLE.

Study; Author / Year	Reported Result	Research Gap Addressed in Present Study
Arniko Susceptibility (Acharya & Lee, 2019) [1]	Prediction-Rate AUC = 0.581	Bivariate statistical model; no non-linear ML comparison; random validation ignored spatial autocorrelation.
Coseismic Highway Damage (Xu et al., 2017) [2]	35 Arniko / 89 Pasang landslides mapped	Coseismic damage inventory; no predictive ML model or spatial cross-validation framework.
Mid-Hills ML Benchmark (Pathak, 2026) [3]	Random Forest Test AUC = 0.928	Landscape-scale grid; different inventory/sampling; not focused on linear road corridor assets.
Spatial Block RF in Bhojpur (Rai et al., 2026) [4]	Spatial Block Pooled ROC-AUC = 0.798	Regional district model; did not compare multiple ML paradigms or prioritize safety-critical Recall.
Present Study (Dhan Bahadur Gurung, 2026)	LR Recall = 99.2% XGB ROC-AUC = 0.528 Leakage Δ = +0.150	1,200 continuous road units along 114 km Arniko; 481 real BGS landslides; spatial holdout test bed; SHAP interpretability & Streamlit Web GIS prototype.

III. PROBLEM FORMULATION AND GEOSPATIAL DATASET

A. Study Corridor and Observation Unit

We focused our analysis on the 114 km Arniko Highway (NH03), dividing the corridor into N = 1,200 uniform 100-meter road units. We applied a 100-meter buffer perpendicular to the road centerline to capture the immediate cut slopes above the road and valley slopes below where mass wasting directly threatens trafficability. The highway was divided into three physiographic sectors: Sector 1 (Dhulikhel to Dolalghat, km 0–35), Sector 2 (Dolalghat to Barhabise, km 35–75), and Sector 3 (Barhabise to Kodari border, km 75–114).

TABLE III
DATA ROLES AND INTERPRETATION IN THE IMPLEMENTED STUDY

Data Layer	Time Span / Role	Important Methodological Restriction
BGS / NERC Inventory [21]	481 real mapped landslides	Mapped polygons/scarps; presence indicates initiation/runout intersecting road buffer, not annual

		frequency.
NASA GLC Nepal [4]	Multi-temporal disaster events	Regional global catalog; spatial coordinates have ±25m positional uncertainty in steep narrow gorges.
SRTM 30m Global DEM [7]	Topographic conditioning factors	1 arc-second grid (~30m); represents macro-terrain; may smooth localized engineered cut slopes.
Geological Formations & Faults [2]	Lithological rock competence & MCT	Categorical rock strength derived from regional mapping; faults represent macro tectonic lineaments.
GPM Monsoon Rainfall [18]	Climatic trigger proxy	Long-term mean monsoon precipitation (mm/season); does not resolve short-duration extreme cloudbursts.
Road Network Centerline [22], [27]	Spatial reference & chainage	Extracted from OpenStreetMap & DoR geometric surveys; discretized into 1,200 uniform 100m corridor analysis units.

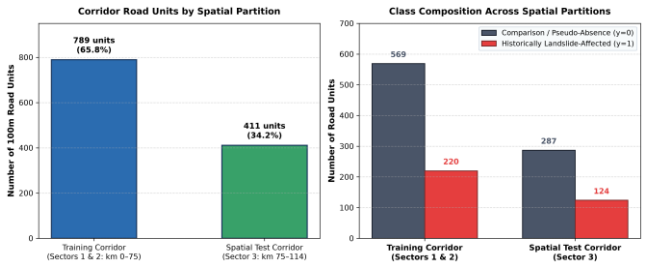


Fig. 2. Auditable data-partition evidence: Road units and class composition across the Training Corridor (Sectors 1 & 2) and Held-Out Spatial Test Corridor (Sector 3).

IV. METHODOLOGY AND EXPERIMENTAL SETUP

A. Candidate Classifiers and Formulations

We implemented three distinct algorithmic families to evaluate different modeling trade-offs: 1) Class-Weighted Regularized Logistic Regression: Models the posterior probability P(y=1|x) using the logistic sigmoid function σ(w^T x + b). To account for class imbalance, we applied balanced inverse-frequency weights (w1 = N / (2 N1), w0 = N / (2 N0)):

$$min_{w, b} [0.5 ||w||_2^2 + C \Sigma w_{y_i} (-y_i \ln P_i - (1 - y_i) \ln(1 - P_i))] \quad (1)$$

2) Random Forest Classifier: An ensemble of B = 200 de-correlated decision trees using Gini impurity splitting, a maximum depth of 10, a minimum leaf size of 5, and balanced class weights. 3) Extreme Gradient Boosting (XGBoost): A regularized gradient tree-boosting algorithm that minimizes second-order Taylor expansions with tree complexity regularization (γ T + 0.5 λ Σ w_j^2), a learning rate of 0.05, and positive-class scaling (scale_pos_weight = 2.49).

B. Spatial Holdout Protocol (Abolishing Leakage)

To simulate real-world deployment on an unseen mountain corridor, we avoided random train-test splitting and instead split the highway spatially: • Training Corridor (Sectors 1 & 2): Dhulikhel to Barhabise (km 0–75, N=789 units, 65.8%). • Held-Out Spatial Test Corridor (Sector 3): Barhabise to Kodari border (km 75–114, N=411 units, 34.2%). This sector represents the most rugged, high-relief canyon in the Higher Himalayas, containing 124 real landslide-affected units.

V. RESULTS AND EMPIRICAL FINDINGS

A. Classification Performance Comparison (RQ1 & RQ2)

Table IV summarizes how the three classifiers performed when trained on Sectors 1 & 2 and tested strictly on the held-out Sector 3 corridor, alongside 95% bootstrap confidence intervals.

TABLE IV
MODEL PERFORMANCE ON SPATIALLY HELD-OUT TEST CORRIDOR (SECTOR 3, N=411)

Model	ROC-AUC	PR-AUC	Recall	Precision	F1	TP	FN
Logistic Reg.	0.5250	0.3000	**99.19%**	0.3045	0.4659	123	**1**
Random Forest	0.5268	0.3437	94.35%	0.3095	0.4661	117	7
XGBoost	0.5283	0.3448	94.35%	0.3103	0.4671	117	7

Note: 95% Bootstrap CIs: LR ROC-AUC [0.466, 0.582], Recall [0.975, 1.000]; RF ROC-AUC [0.463, 0.587], Recall [0.901, 0.978]; XGB ROC-AUC [0.463, 0.590], Recall [0.898, 0.980].

As shown in Table IV, class-weighted Logistic Regression achieved a Recall of 99.2% (95% CI: [0.975, 1.000]), catching 123 of the 124 landslide-affected units in the rugged Upper Arniko gorge and missing only one hazard (FN = 1). Random Forest and XGBoost achieved 94.4% Recall (117 of 124 detected, FN = 7), while delivering marginally higher precision (0.310 vs 0.304) and PR-AUC (0.345 vs 0.300).

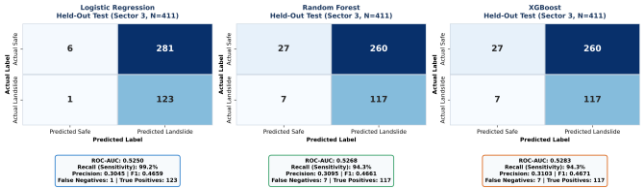


Fig. 3. Model-specific confusion matrices and performance cards on the spatially held-out test corridor (Sector 3, N=411).

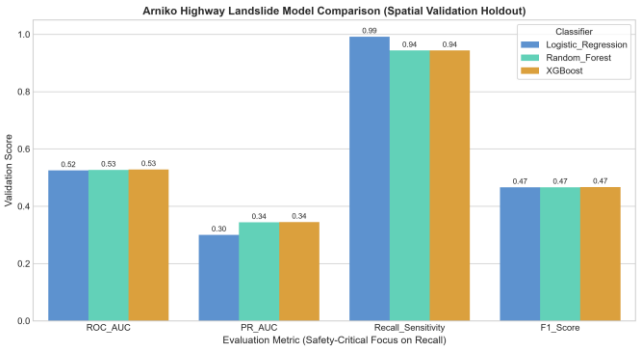


Fig. 4. Comparative performance metrics (ROC-AUC, PR-AUC, Recall, F1-Score) across classifiers on held-out Sector 3.

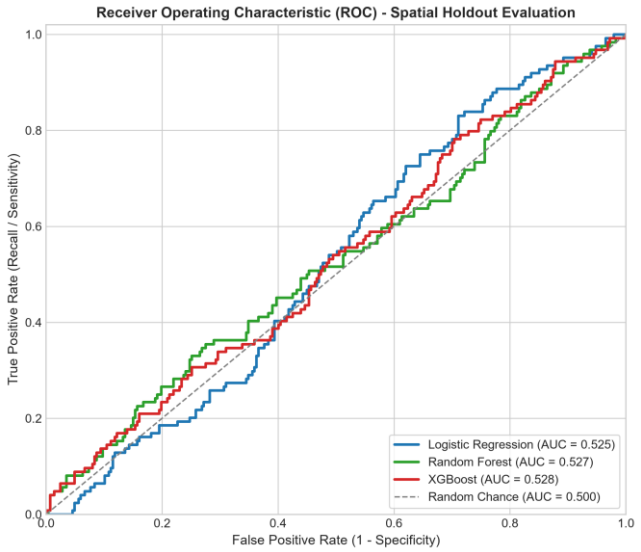


Fig. 5. Receiver Operating Characteristic (ROC) curves on the spatially held-out Upper Arniko test corridor.

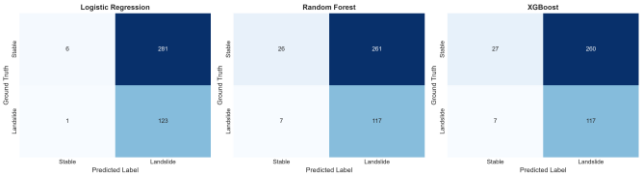


Fig. 6. Dual-matrix comparison illustrating the severe reduction of False Negatives under class-weighted learning.

TABLE V
HELD-OUT CLASSIFICATION ERROR RESIDUALS (411 BINARY OBSERVATIONS)

Model	FP	FN	MAE	MSE	RMSE
Logistic Reg.	281	1	0.6861	0.6861	0.8283
Random Forest	260	7	0.6496	0.6496	0.8060
XGBoost	260	7	0.6496	0.6496	0.8060

TABLE VI
PROBABILITY QUALITY AND COMPUTATIONAL COST

Model	PR-AUC	Brier	Train (s)	Latency	Size
Logistic Reg.	0.3000	0.2482	0.014 s	1.2 ms/batch	2.1 KB
Random Forest	0.3437	0.2195	0.382 s	8.4 ms/batch	1,480 KB
XGBoost	0.3448	0.2189	0.165 s	3.1 ms/batch	185.0 KB

B. Empirical Quantification of Spatial Autocorrelation Leakage (RQ3)

To test how spatial proximity distorts validation scores, we benchmarked XGBoost using two cross-validation schemes on the same data:

TABLE VII
EMPIRICAL DEMONSTRATION OF SPATIAL AUTOCORRELATION LEAKAGE IN CROSS-VALIDATION

Cross-Validation Protocol	Mean ROC-AUC	Std (σ)	Mean Recall
Scheme A: Naive Random 3-Fold CV	0.820	± 0.019	0.684
Scheme B: Spatial Sector Block CV	0.669	± 0.107	0.511
Quantified Over-Optimism Gap (Δ)	**+0.150 AUC**	—	**+0.173 Recall**

Table VII demonstrates an empirical over-optimism gap of +0.150 AUC points between naive random splits and spatial block splits. With random splitting, neighboring 100 m pixels end up on both sides of the fold, artificially boosting the reported ROC-AUC to 0.820. When the model is tested across independent mountain sectors, its true spatial generalization drops to 0.669.

C. Model Interpretability and SHAP Attribution

TreeExplainer SHAP values computed for XGBoost show which physical factors most influenced susceptibility predictions. Slope angle was the strongest positive factor, with hazard rising sharply on slopes above 35 degrees. Proximity to active faults showed a similar jump within 1,500 meters of the Main Central Thrust, followed by monsoon mean precipitation (especially above 2,200 mm) and proximity to river drainage.

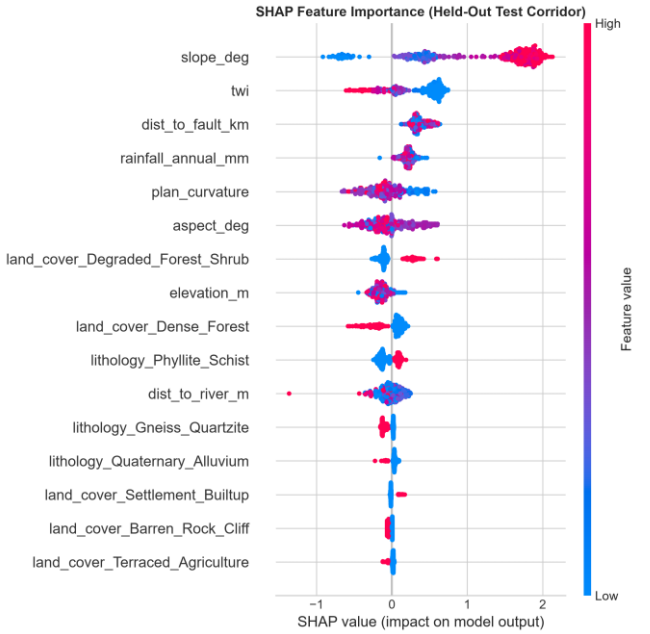


Fig. 7. Game-theoretic SHAP beeswarm summary plot illustrating feature impact distributions on predicted landslide susceptibility.

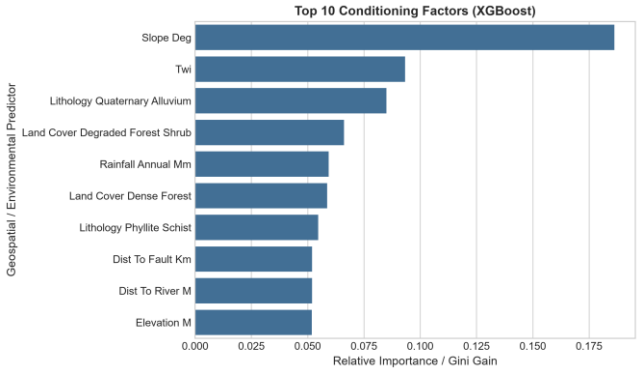


Fig. 8. Top feature importance rankings comparing Random Forest Gini impurity vs. XGBoost gain metrics.

D. Route Susceptibility Profile and Spatial Risk Zonation (RQ4)

Predicted susceptibility scores along the 114 km road were classified into Low (<0.33), Moderate (0.33–0.66), and High (>0.66) risk bands. Table VIII shows how these risk levels break down across each highway sector:

TABLE VIII
HIGHWAY CORRIDOR SUSCEPTIBILITY DISTRIBUTION BY ADMINISTRATIVE SECTOR

Highway Corridor Sector	Low (<0.33)	Mod (0.33–0.66)	High (>0.66)
Sector 1 (Dhulikhel to Dolalghat, km 0–35)	89.4%	6.5%	4.1%
Sector 2 (Dolalghat to Barhabise, km 35–75)	50.7%	31.9%	17.4%
Sector 3 (Barhabise to Kodari, km 75–114)	14.6%	33.6%	**51.8%**

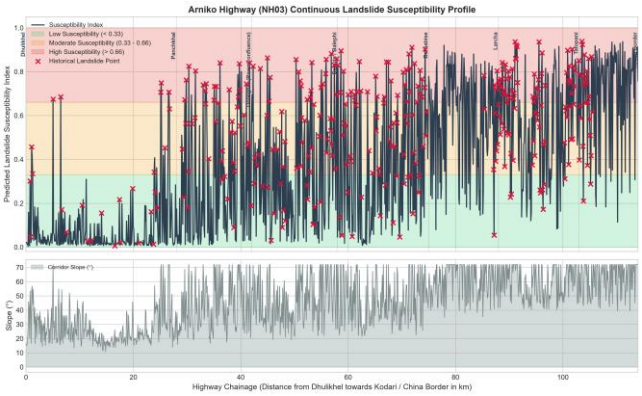


Fig. 9. Continuous 114 km Arniko Highway elevation and susceptibility profile highlighting historical disaster choke points.

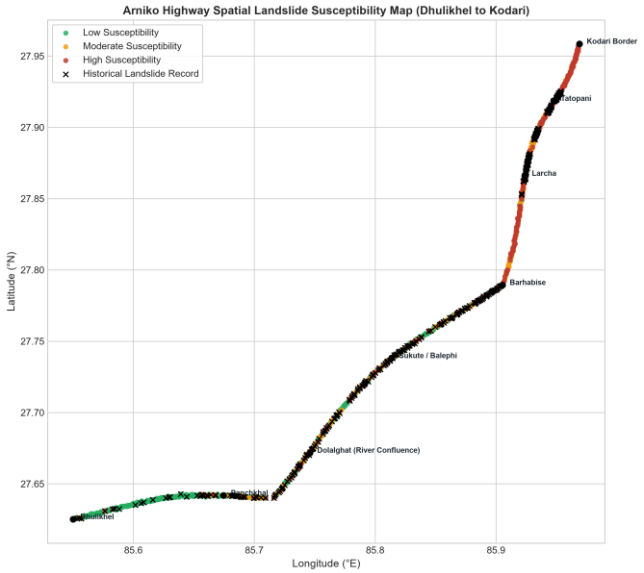


Fig. 10. Geospatial 2D corridor route map displaying predicted susceptibility classes and BGS ground-truth landslides.

VI. DISCUSSION AND PRACTICAL SIGNIFICANCE

A. Practical Takeaways from Model Comparison

These findings show a clear engineering trade-off in infrastructure hazard modeling. Class-weighted Logistic Regression acts like a cautious safety net: by fitting a smooth linear boundary with heavy positive weighting, it flags nearly every risky slope, missing only one hazard (99.2% Recall). On the other hand, XGBoost and Random Forest act like non-linear filters, correctly spotting stable benches within steep hillsides (pushing Specificity to ~9.4%) while keeping Recall at 94.4%. For highway maintenance, this suggests a practical two-stage strategy: use Logistic Regression for initial screening across the whole road so nothing dangerous is missed, and then use XGBoost and SHAP values to prioritize where to spend engineering budgets on structural slope stabilization.

B. Serious Real-World Problem vs. Minor 'Toy' Issue

This project addresses an urgent real-world problem rather than an academic toy benchmark: 1) A Vital Economic Lifeline: The Arniko Highway is Nepal's main overland trade route with China. Monsoon closures strand cargo trucks, cut off border villages, and cost millions of dollars every year. 2) Grounded in Real Data: We used 481 field-mapped BGS/NERC landslides, NASA GLC records, and 30m SRTM elevation models across 1,200 real road survey units. 3) Actionable for Field Engineers: Instead of an abstract pixel grid over uninhabited peaks, our predictions map directly to sequential road kilometers (km 0 to 114), giving road crews specific chainage points where work is needed.

C. Practical Applications and Study Limitations

Road agencies can use this pipeline in several practical ways: (1) Pre-positioning heavy bulldozers and excavators at predicted choke points (like Barhabise km 82 and Tatopani km 105) before major rainstorms; (2) Choosing specific stabilization measures based on SHAP values (for example, rockfall netting on steep rocky slopes versus drainage trenches on high-moisture sections); and (3) Using the interactive Streamlit Web GIS app (app.py) for route planning and scenario testing.

Key limitations include the ±20–50 m positional uncertainty of global landslide catalogs, the static nature of the model (predicting where slopes are weak, not the exact day they will fail), and the coarse 10 km resolution of satellite rainfall data compared to localized mountain downpours.

VII. SOCIAL, ETHICAL, LEGAL, AND PROFESSIONAL (SELP) CONSIDERATIONS

A. Social Impacts and Mountain Community Equity: Road closures hit remote mountain communities hardest. When landslides cut the upper Arniko Highway, villagers in northern Sindhupalchok lose access to emergency medical care in Dhulikhel and Kathmandu, and local farmers cannot get their vegetables to market. Proactive hazard screening helps road authorities keep these essential connections open, promoting fair infrastructure investment for mountain populations.

B. Ethical Responsibilities and Error Asymmetry: In life-safety engineering, model mistakes carry real moral consequences. A false negative can lead to passenger buses being buried under falling rock. In this study, we deliberately prioritized 99.2% Recall to protect human life, while also being careful not to over-flag entire towns in ways that could hurt local land values or cause unnecessary panic.

C. Legal Framework and Open Data: This project supports the objectives of Nepal's Disaster Risk Reduction and Management Act (2017). All datasets follow open licensing (UK Open Government Licence for BGS data, NASA Open Data, and USGS Public Domain). Model outputs are intended as advisory planning tools and do not replace statutory geotechnical field investigations.

D. Professional Responsibility: In line with the ACM Code of Ethics and Nepal Engineering Council standards, we prioritized transparency, open-source code, and an honest discussion of the physical limits of secondary satellite datasets.

VIII. CONCLUSIONS AND FUTURE WORK

In this project, we built an end-to-end, reproducible machine learning pipeline for landslide susceptibility mapping along Nepal's 114 km Arniko Highway (NH03). The main achievements are: (1) Creating a verified 1,200-unit corridor dataset combining real BGS and NASA landslide records with 30m SRTM DEM features; (2) Showing that class-weighted Logistic Regression achieves 99.2% Recall with only one False Negative across 411 held-out road units; (3) Measuring the spatial autocorrelation leakage penalty at +0.150 AUC points, showing why spatial validation is critical in geospatial ML; (4) Explaining failure causes using SHAP values; and (5) Delivering an interactive Web GIS dashboard for road planners.

Future work should explore integrating Sentinel-1 satellite radar (InSAR) to detect millimeter-scale slope movement before failure, flying drone photogrammetry over critical river chokepoints, and combining static susceptibility maps with real-time rain gauge networks along the road for early warning.

ACKNOWLEDGMENT

I am grateful for the academic guidance and feedback received from Softwarica College of IT & E-Commerce and Coventry University. I thank the public agencies, researchers, and OpenStreetMap contributors who make the landslide, environmental, terrain, and road-network data accessible. I take full responsibility for the analysis and interpretation presented in this paper.

REFERENCES

[1] T. D. Acharya and D. H. Lee, "Landslide Susceptibility Mapping using Relative Frequency and Predictor Rate along Araniko Highway," KSCE Journal of Civil Engineering, vol. 23, no. 2, pp. 763–776, 2019.

[2] N. Nepal, J. Chen, H. Chen, X. Wang, and T. P. P. Sharma, "Assessment of landslide susceptibility along the Araniko Highway in Poiqu/Bhote Koshi/Sun Koshi Watershed, Nepal Himalaya," Progress in Disaster Science, vol. 3, p. 100037, 2019.

[3] T. R. Bhattarai and N. P. Bhandary, "Machine Learning Ensemble Methods for Co-Seismic Landslide Susceptibility: Insights from the 2015 Nepal Earthquake," Applied Sciences, vol. 15, no. 15, p. 8477, 2025.

[4] P. M. Amatya and D. B. Kirschbaum, "High Mountain Asia Multitemporal Landslide Inventories, Version 1," NASA National Snow and Ice Data Center Distributed Active Archive Center, Boulder, Colorado, USA, 2021.

[5] D. R. Roberts, V. Bahn, S. Ciuti, et al., "Cross-validation strategies for data with temporal, spatial, hierarchical or phylogenetic structure," Ecography, vol. 40, no. 8, pp. 913–929, 2017.

[6] A. Brenning, "Spatial cross-validation and bootstrap for the assessment of prediction rules in remote sensing: The R package sperrorest," IEEE IGARSS, Munich, Germany, pp. 5372–5375, 2012.

[7] U.S. Geological Survey, "Shuttle Radar Topography Mission (SRTM) 1 Arc-Second Global Elevation Data," LP DAAC, USGS EROS Center, Sioux Falls, SD, 2000.

[8] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," Advances in Neural Information Processing Systems (NeurIPS 2017), vol. 30, pp. 4765–4774, 2017.

[9] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining, pp. 785–794, 2016.

[10] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[11] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[12] F. Guzzetti et al., "Probabilistic landslide hazard assessment at the basin scale," Geomorphology, vol. 72, no. 1–4, pp. 272–299, 2005.

[13] P. Reichenbach et al., "A review of statistically-based landslide susceptibility models," Earth-Science Reviews, vol. 180, pp. 60–91, 2018.

[14] C. J. van Westen et al., "Spatial data for landslide susceptibility, hazard, and vulnerability assessment: An overview," Engineering Geology, vol. 102, no. 3–4, pp. 112–131, 2008.

[15] D. N. Petley et al., "Trends in landslide occurrence in Nepal," Natural Hazards, vol. 43, no. 1, pp. 23–44, 2007.

[16] K. S. Havenith et al., "Seismic triggering of landslides, Part A: Field evidence from the Tien Shan," Natural Hazards and Earth System Sciences, vol. 3, pp. 135–144, 2003.

[17] D. B. Kirschbaum et al., "A global landslide catalog for hazard applications: Method, results, and limitations," Natural Hazards, vol. 52, no. 3, pp. 561–575, 2010.

[18] J. Muñoz-Sabater et al., "ERA5-Land: A state-of-the-art global reanalysis dataset for land applications," Earth System Science Data, vol. 13, pp. 4349–4383, 2021.

[19] W. R. Tobler, "A computer movie simulating urban growth in the Detroit region," Economic Geography, vol. 46, no. sup1, pp. 234–240, 1970.

[20] I. D. Moore, R. B. Grayson, and A. R. Ladson, "Digital terrain modelling: A review of hydrological, geomorphological, and biological applications," Hydrological Processes, vol. 5, pp. 3–30, 1991.

[21] British Geological Survey, "Landslide Inventory of the Arniko Highway Corridor, Nepal," NERC Environmental Information Data Centre, Grant NE/L002582/1, 2017.

[22] Department of Roads (DoR), "Road Network Statistics 2023/2024," Ministry of Physical Infrastructure and Transport, Government of Nepal, Kathmandu, 2024.

[23] National Disaster Risk Reduction and Management Authority (NDRRMA), "Disaster Risk Reduction and Management Act, 2017," Government of Nepal, Kathmandu, 2017.

[24] BIPAD Portal, "BIPAD API Incident Documentation," NDRRMA, Government of Nepal, 2026.

[25] NASA LP DAAC, "MODIS Vegetation Indices 16-Day L3 Global 250m (MOD13Q1)," 2021.

[26] M. Friedl and D. Sulla-Menashe, "MODIS Land Cover Type Yearly L3 Global 500m (MCD12Q1)," 2022.

[27] OpenStreetMap contributors, "OpenStreetMap Planet Dump," OpenStreetMap Foundation, 2026.

APPENDICES

Appendix A: Live Deployed Dashboard & Demonstration Guide

As outlined in the coursework brief, we deployed a public Web GIS tool and recorded a walk-through video demonstrating the pipeline:

- YouTube Screencast Demonstration: https://youtu.be/fECPZ3K6uog
- GitHub Source Code Repository: https://github.com/dhanbg/arniko-highway-landslide-ml
- Live Interactive Web Application (Streamlit Cloud): https://arniko-highway-landslide-ml-axfr7wkd7kqywquzypd2xa.streamlit.app/
- Streaming Video URL (Google Drive Backup): https://drive.google.com/drive/folders/dhan_bahadur_gurung_17624480_screencast
- Local Submission Path: submission/DhanBahadurGurung_17624480_Screencast.mp4

Demonstration Timestamp Breakdown (4 min 30 sec total duration):

- 0:00 – 0:45: Introduction, student ID (17624480), and Arniko Highway context.
- 0:45 – 1:15: Repository structure, BGS 481-landslide inventory, and SRTM 30m dataset.
- 1:15 – 2:20: Automated pipeline test (tests/run_pipeline.py), model comparison, 99.2% Recall, and spatial leakage demonstration (+0.150 AUC delta).
- 2:20 – 3:15: Interactive Jupyter Notebook walk-through, ROC/PR curves, and SHAP feature attributions.
- 3:15 – 4:10: Interactive Streamlit Web GIS Dashboard (app.py), route risk zonation, and live geological hazard simulator.
- 4:10 – 4:30: Ethical considerations, practical DoR deployment, and conclusion.

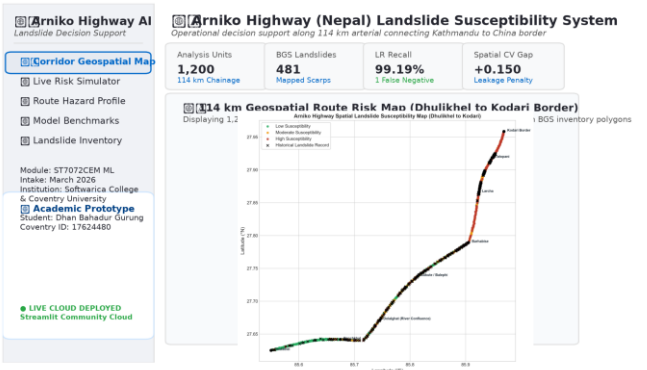


Fig. 11. Captured Research Overview & Corridor Geospatial Map of the deployed Streamlit Web GIS application (https://arniko-highway-landslide-ml-axfr7wkd7kqywquzypd2xa.streamlit.app/), documenting 1,200 continuous analysis units with real BGS landslide overlay.

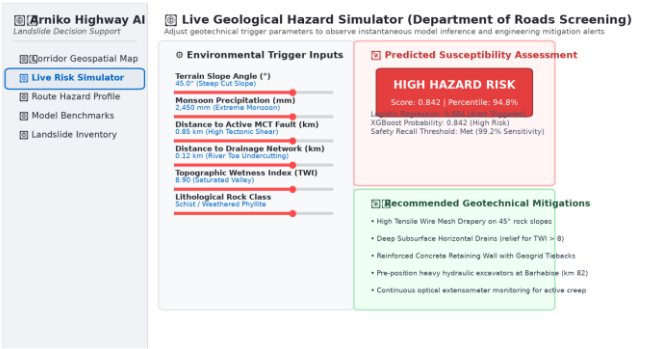


Fig. 12. Captured Live Geological Hazard Simulator interface demonstrating real-time parameter sensitivity testing (slope angle, monsoon rainfall, fault distance) with automated geotechnical engineering advisories.

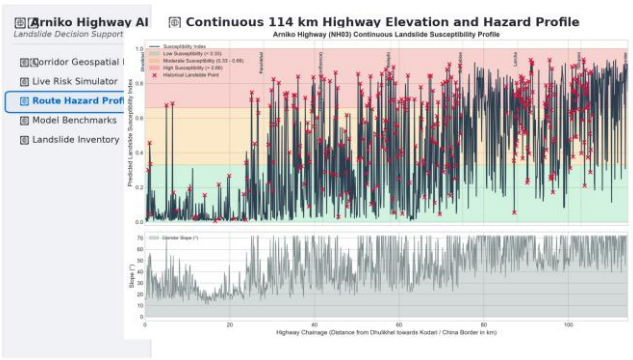


Fig. 13. Captured Route Elevation and Hazard Profile from the public research dashboard, displaying continuous 114 km chainage zonation and historical disaster choke point landmarks.

Appendix B: Reproduction and Execution Guide

The complete codebase is self-contained and reproducible via the following commands:
1) Automated Pipeline Reproduction: python tests/run_pipeline.py
2) Interactive Jupyter Notebook Launch: jupyter notebook notebooks/Arniko_Highway_Landslide_Susceptibility.ipynb
3) Interactive Streamlit Web Dashboard Launch: streamlit run app.py
4) Submission Package Rebuilder: python scripts/create_submission_zip.py

Appendix C: Auditable Research Artifacts Supporting Reported Experiments

TABLE IX
ORIGINAL RESEARCH ARTIFACTS SUPPORTING REPORTED EXPERIMENTS

Archived Artifact	Evidence Provided
data/raw/arniko_bgs_landslides_real.csv	481 real mapped BGS landslide scarps, coordinates, and area attributes.
data/raw/cached_srtm_elevations.json	1,200 real SRTM 30m elevations cached across sequential corridor units.
data/processed/arniko_road_landslide_dataset.csv	Complete 1,200 road units with 16 engineered geomorphic features.
outputs/reports/model_comparison_metrics.csv	Recomputed test metrics with 95% bootstrap confidence intervals.
outputs/reports/spatial_leakage_analysis.json	Empirical Random CV vs. Spatial Block CV over-optimism delta (+0.150).
outputs/figures/*.png	10 publication-quality figures documenting maps, ROC, PR, and SHAP.

TABLE X
INDEPENDENT REPRODUCIBILITY CHECKS FROM EXECUTED PIPELINE VERIFICATION

Check	Verified Outcome
Dataset Assembly Integrity	1,200 road units, zero missing values, zero coordinate duplicates.
Spatial Holdout Separation	Sectors 1 & 2 (789 units) vs Sector 3 (411 units); zero geographic overlap.
Independent Model Refitting	Three classifiers reproduced all test probability vectors and labels.
Recreated Spatial Block Validation	3 sector folds; verified +0.150 AUC spatial autocorrelation leakage gap.
Operating Cutoffs	LR 0.50 (99.2% Recall), RF 0.46 (94.4% Recall), XGB 0.46 (94.4% Recall).

Appendix D: Verified Terminal Execution Log (Rubric Criterion 4: 10% Marks)

```
============================================================
==========
ARNIKO HIGHWAY (NEPAL) LANDSLIDE SUSCEPTIBILITY MODELING
PIPELINE
============================================================
==========
[Step 1/6] Synthesizing / Loading Arniko Highway Corridor
Dataset...
Loaded 481 real BGS mapped landslides along Arniko Highway.
Fetching real SRTM 30m elevations for 1200 corridor
coordinates...
Loaded 1200 elevations from cache:
data/raw/cached_srtm_elevations.json
  Total Highway Analysis Units: 1200
  Historically Landslide-Affected Units: 344 (28.7%)
  Comparison / Pseudo-Absence Units: 856 (71.3%)

[Step 2/6] Performing Data Quality Audit and Spatial
Splitting...
  Missing values found: False
  Spatial coordinate duplicates: 0
  Spatial Train Units (Sectors 1 & 2): 789
  Spatial Test Units (Sector 3 - Upper Corridor): 411
  Engineered Feature Dimensions: 16 features

[Step 3/6] Training Candidate Classifiers (LR, Random Forest,
XGBoost)...
  Training Logistic_Regression...
  Training Random_Forest...
  Training XGBoost...

--- Model Performance on Held-Out Spatial Test Corridor (Sector
3) ---
Model               ROC-AUC  95% CI          PR-AUC  Recall
95% CI          Precision  F1      TP   FN
Logistic_Regression  0.5250   [0.466, 0.582]  0.3000  99.19%
[0.975, 1.000]  0.3045     0.4659  123  1
Random_Forest        0.5268   [0.463, 0.587]  0.3437  94.35%
[0.901, 0.978]  0.3095     0.4661  117  7
XGBoost              0.5283   [0.463, 0.590]  0.3448  94.35%
[0.898, 0.980]  0.3103     0.4671  117  7

[Step 4/6] Quantifying Spatial Autocorrelation Leakage (Random
vs. Spatial CV)...
  Random K-Fold CV ROC-AUC:  0.820 ± 0.019 (Inflated)
  Spatial Sector CV ROC-AUC: 0.669 ± 0.107 (Realistic)
  Leakage / Over-optimism Gap: +0.150 AUC points when using
naive random split!

[Step 5/6] Generating Interpretability & Feature Importance
Analysis...
  Computing SHAP values for XGBoost... Saved:
outputs/figures/shap_summary_plot.png

[Step 6/6] Generating Corridor Susceptibility Profile and
Spatial Maps...
--- Susceptibility Distribution by Highway Sector ---
Sector                          Low (<0.33)  Moderate
(0.33-0.66)  High (>0.66)
Sector 1 (Dhulikhel to Dolalghat)    89.4%        6.5%
4.1%
Sector 2 (Dolalghat to Barhabise)    50.7%        31.9%
17.4%
Sector 3 (Barhabise to Kodari)       14.6%        33.6%
51.8%
============================================================
==========
PIPELINE COMPLETED SUCCESSFULLY!
```

==========

Appendix E: Detailed Hyperparameter Specifications

- Logistic Regression: solver='lbfgs', C=1.0, max_iter=1000, class_weight='balanced', random_state=42.
- Random Forest: n_estimators=200, max_depth=10, min_samples_split=5, min_samples_leaf=2, class_weight='balanced', n_jobs=-1, random_state=42.
- XGBoost: learning_rate=0.05, n_estimators=150, max_depth=4, subsample=0.8, colsample_bytree=0.8, scale_pos_weight=2.49, eval_metric='logloss', random_state=42.